

MOHAMED ZAYED AHMED

+201284907633 moh.z.ahmed007@gmail.com linkedin.com/in/mozayed007 github.com/mozayed007

EXPERIENCE

Sprints AI

Research Engineer - R&D

Cairo, EG

Sep 2025 – Present

- Owned end-to-end architecture and delivery of the Learning Companion PoC; worked with R&D engineering team to productionize the stack.
- Productionized HelixDB v2's graph+vector hybrid retrieval (native traversals) within the Learning Companion, achieving 50% fewer DB round-trips and superior curriculum-aware relevance.
- Built a dual RAG provider adapter (Qdrant ↔ HelixDB) with runtime switching and graceful fallback, enabling zero-downtime provider changes.

Freelance Consulting & Contracts

Remote

ML Engineer (part-time) - (selected projects alongside full-time roles)

Sep 2023 – Present

- Multi-agent research framework:** Architected and implemented a dynamic, DeepResearch-inspired multi-agent system using MCP servers to orchestrate autonomous research workflows.
- Autonomous NER pipeline:** Designed and built a responsible, end-to-end NER pipeline for unstructured data, featuring continuous model fine-tuning and automated data ingestion.
- ML-driven pharmaceutical granulation:** Engineered an optimized ODT model to replace a legacy granulation controller (MPC), focusing on model interpretability and ethical innovation in healthcare AI.
- Explainable dynamic modeling:** Passionately designed scalable control strategies for life science processes under uncertainty using Bayesian methods (MOBO, GP, BART) and generative models (CVAE).
- LLM Evaluation System (for Stakpak):** Through effective collaboration, implemented a multi-processed async benchmark system to evaluate and optimize LLMs, boosting experimentation speed by 3x (from 30-50 minutes to 10 minutes for the generation/evaluation pipeline) and ensuring provider interoperability.

Proteinea

Deep Learning Intern

Cairo, EG

Aug 2022 – Nov 2022

- Developed initial T5-based models for antigen-antibody interactions, laying foundation for graduation thesis that later surpassed SoTA baselines.
- Automated evaluation pipelines and created TensorBoard visualizations, reducing analysis time by 30%.

Superprof

Online Tutor · Freelance

Remote, US

Aug 2022 – Present

- Tutored undergraduate (B.Sc) and graduate (M.Sc) students in Machine Learning, Deep Learning, NLP, Computer Vision, and Cybersecurity.
- Completed 250+ hours of personalized online tutoring, improving student performance on projects and advanced SOTA topics.

PROJECTS

Antigen-Antibody Translation/Generation – PyTorch, HuggingFace, T5, Pandas, AWS

Graduation Research Thesis

- Developed a custom T5 Transformer that achieved 10% higher performance than 2022 SoTA baselines on the SABDAB dataset (building on insights from Proteinea internship) using AWS NVIDIA A10G GPUs.
- Enabled real-time monitoring with TensorBoard dashboards and facilitated team collaboration using Termius Vault and Tmux.

Multi-Agent Research System – Google ADK, Pydantic, asyncio, FastAPI, Gradio

- Drove innovation by migrating a research workflow to Google's Agent Development Kit (ADK), creating a SequentialAgent pipeline for data collection, analysis, and report synthesis.
- Integrated external APIs (BraveSearch, ArXiv, Wikipedia) and SQLite persistence to enhance system modularity and data gathering capabilities.
- Drafted architecture diagrams and communication patterns to streamline development and ensure scalability.

SKILLS

Programming Languages: Python (learning Triton/CuTe), C++ (learning CUDA).

Frameworks & Libraries: PyTorch, TensorFlow, FastAPI, LitServe, Triton, Pandas (CuDF), NumPy (CuPy), Hugging Face (Transformers, PEFT), Scikit-Learn, Timm, OpenCV, NLTK, LangChain, LlamaIndex, Pydantic

Domain skills: Machine Learning, Deep Learning, NLP (NER, LLMs, RAG), Computer Vision, Generative AI, Statistical Inference, Data Analysis, Vector Databases (Milvus, Qdrant, Weaviate)

Tools & Platforms: Git, Docker, AWS, GCP, Azure, Ollama, Linux, WSL, SQLite, REST APIs, Kaggle

EDUCATION

University of Science and Technology – Zewail City

Bachelor of Science, Communication and Information Engineering

Giza, EG

Jun 2024